

Saam Haghighat-Grami

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Roswell, GA

Interests – Autonomous, Mobile, Collaborative Robotics

Graduating December 2026

LinkedIn – saam-haghighat-grami

GitHub – Saam-Grami

Portfolio – saam-grami.github.io

Active Military Secret Security Clearance

EDUCATION

• Kennesaw State University

Mechatronics Engineering, B.S. – Graduation December 2026

January 2022 – May 2022, January 2024 – Present

CGPA: 4.00

• Marine Corps Communication Electronics School

Ground Electronics Transmission Systems Maintainer

August 2022 – October 2023

CGPA: 4.00

TECHNICAL SKILLS

Autonomy & Robotics: ROS 2, Gazebo, Nav2, ArduPilot, PX4, Mission Planner, UAV/UGV integration, RTAB Map

Embedded & Hardware: ESP32, Arduino, Raspberry Pi 5, Jetson Orin Nano, OpenMV, LiDAR, RGB-D cameras, IMUs, ad hoc networking, Control Theory

Languages & Tools: C++, Python, MATLAB, LaTeX, Linux/WSL2, Git, VS Code, SolidWorks, Onshape

Perception & ML: PyTorch, TensorFlow, YOLO, CNNs, LSTMs, OpenCV, ORB-SLAM3, sensor fusion

Manipulators & Automation: Denavit-Hartenberg parameters, forward and inverse kinematics, trajectory planning, Ladder Logic, TIA Portal, PLCs, HMI development

 **Portfolio** — Additional projects, write-ups, and technical details available at saam-grami.github.io

RESEARCH & ENGINEERING EXPERIENCE

• MEAP – Mass Estimation Autonomous Pathfinding for Search and Rescue

November 2025 - Present

Robotics Software Architect & Project Coordinator – AgriSys Lab, Dr. Muhammad Hassan Tanveer

- **Overview:** UAV-UGV collaborative search-and-rescue system: the UAV stitches aerial maps while segmenting and identifying objects below; the UGV traverses via SLAM and fuses its sensor data with the UAV's to layer maps and estimate each object's volume and mass, determining whether it can be moved to shorten the path to the objective.
- **Contribution:** Leading a 5-person team on overall planning and execution; built the UAV autonomy stack (flight motion framework, aerial map stitching), validated on a 2 m² test area and scaling toward 25 m². Also leading the ML pipeline that estimates obstacle mass from sensor-fused SLAM point clouds, targeting 85% mission success with the manipulator handling obstacles up to ~0.5 kg.

• DARB – Drone Arm Robotic Backpack (Personal Project)

May 2025 - Present

Sole Designer and Builder

- **Overview:** Autonomous drone docking system: a backpack-mounted robotic arm captures a drone mid-flight, stows it, and relaunches it, targeting a 30-second capture-to-relaunch cycle from stable hover.
- **Approach:** Custom airframe uses computer vision and machine learning to align to IR LED markers, triggering the onboard-controlled capture arm once orientation converges.
- **Hardware:** Matek H743 flight controller (ArduPilot), encoder-driven NEMA 23 steppers actuating the capture arm.

• Trajectory Forecasting for Floating Debris in Waterways

June 2026 - August 2026

Funded Undergraduate Researcher – Summer Undergraduate Research Program, Dr. Matt Marshall

- **Overview:** Real-time vision system detecting, tracking, and forecasting floating river debris trajectories from a fixed riverbank camera, directing a collection platform to intercept debris.
- **Approach:** Fuses dense optical-flow surface velocimetry with object motion/slip in a CNN + LSTM-DNN forecaster; Gaussian head yields calibrated multi-step forecasts over YOLO26 detections, with an object manager reconciling identity across lost tracks.

• SoilBus – Soil Monitoring Probe Network

August 2025 - December 2025

Lead Software Developer – AgriSys Lab, Dr. Muhammad Hassan Tanveer

- **Overview:** Hub-less soil-monitoring mesh network: probes auto-discover neighbors, and any single probe can offload the whole network's data.
- **Results:** Full connectivity in 15s; each node offloads a day's sensor data in ~5s. BFS-based inter-probe routing, validated on 5 physical nodes, holding to 26 simulated before degradation.

PUBLICATIONS

- *SoilBus: An Ad Hoc Soil Sensor Network for Precision Agriculture*. Published, IEEE SoutheastCon 2026, Huntsville, AL.

DOI: 10.1109/SoutheastCon63549.2026.11476560 – added to IEEE Xplore April 2026.

- *Mass Estimation Autonomous Pathfinding with Collaborative UGV and UAV*. First-author manuscript in preparation – targeting IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2027.

- *Trajectory Forecasting for Floating Debris in Waterways*. First-author manuscript in preparation – targeting IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2027.

INDUSTRY & MILITARY EXPERIENCE

- **EOSYS Industrial and Manufacturing System Integrator** May 2025 – August 2025
COOP Marietta GA
 - **HMI tooling:** Built a Human Machine Interface display generator that produced over 40 P&ID screens for plant operators.
 - **Controls:** Reviewed and modified PLC programs across two production shops plus the PLC trainer system.
- **United States Marine Corps Reserve** August 2022 - Present
Ground Radio Repair Specialist Marietta GA
 - **Radio systems:** Troubleshoot, maintain, and operate USMC radio systems in time-sensitive conditions; active Secret security clearance.
 - **Squad Leader:** Squad Leader of 8 Marines in the 4th Combat Readiness Regiment, responsible for training, accountability, and readiness.

UPCOMING & IN-PROGRESS PROJECTS

- **Multi-Drone Swarm Firefighting Simulation (Personal Project)** 2026 - Present
Sole Developer
 - **Overview:** PX4/ROS2 multi-drone swarm simulating collaborative wildfire response; progressing through offboard control, sensor fusion, and SLAM/Nav2 toward full swarm coordination.
 - **Current stage:** Implementing namespaced multi-drone control (independent PX4 SITL instances, per-drone offboard nodes) in Gazebo.
 - **Next:** Migrating to NVIDIA Isaac Sim (via Pegasus Simulator) for higher-fidelity swarm simulation at scale.
- **GPS-Denied UAV–UGV Collaborative Localization** 2026 - Present
Lead Researcher – AgriSys Lab, Dr. Muhammad Hassan Tanveer
 - **Overview:** UAV–UGV system propagating GPS error from a moving aerial anchor into an absolute ground-vehicle position estimate in GNSS-denied environments (Clearpath Husky UGV, Pixhawk 2.4.8/PX4 anchor UAV).
 - **Approach:** Fuses UWB range measurements between anchor and UGV with each platform's pose estimate to maintain an absolute reference frame without ground-vehicle GPS.
 - **Status:** Narrowing scope against existing anchor-free GNSS-denied approaches to sharpen the GPS-relay contribution; sourcing a UWB module before hardware validation.